Air–Ground Coordinated MEC: Joint Task, Time Allocation and Trajectory Design

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***Abstract*—The incorporation of uncrewed aerial vehicles (UAVs) presents a promising shift of paradigm for traditional mobile edge computing (MEC) with ground infrastructure. An air-ground coordinated MEC framework provides extensive signaling coverage and additional computing power for user equipments (UEs), especially in remote areas or emerging scenarios. This paper proposes a novel air-ground coordinated MEC system, comprising a laser-powered UAV and a ground access point (AP). In this setup, the UAV serves both as an MEC server and a relay, while the AP supplies energy to the UAV through laser charging. Given the uncertainties in wireless energy harvesting, our goal is to minimize the UAV's long-term average energy consumption by collaboratively optimizing its trajectory, task allocation between the UAV and AP, and time for energy harvesting. To tackle this non-convex optimization problem, we introduce a two-step alternating optimization algorithm that decouples the optimization variables into two subproblems. Subproblem 1 focuses on computation task and energy harvesting time allocation, formulated as a linear programming (LP) problem. Subproblem 2 addresses real-time trajectory scheduling for the UAV, tackled using the deep deterministic policy gradient (DDPG) algorithm. Simulation results demonstrate the algorithm's convergence and its effectiveness in reducing UAV energy consumption compared to benchmark schemes across various scenarios. Additionally, the optimal policies pertaining to UAV trajectory and task allocation ratios are identified and discussed, highlighting key features and implications.**

***Index Terms*—Computation task allocation, energy harvesting time allocation, mobile edge computing (MEC), trajectory design, uncrewed aerial vehicle (UAV), wireless power transfer.**

I. Introduction

RECENT development of wireless communication technologies, such as 5G and the upcoming 6G, have facilitated unprecedented advancement and popularization of highly smart applications, e.g. intelligent navigation, face recognition, voice assistant, and augmented / virtual reality (AR/VR) [1], etc., most of which exhibit computation-intensive and delay-sensitive characteristics [2], yet UEs are typically limited in computing power and battery capacities, while cloud sever brings large delays. Such a dilemma has greatly impeded the quality of experience (QoE) [1] for UEs.

Mobile edge computing (MEC) is recognized as a promising option over local and cloud computing in both academia and industry. MEC relocates mobile computing, network control, and storage away from the cloud and data center, moving them to the edge of the mobile network, i.e., a cellular base station (BS) or access point (AP) [3], [4], making it possible to effectively meet the communication and computing needs of UEs with low latency. However, UEs may be situated in maritime or mountainous regions characterized by sparse distribution and fixed location of BS. UEs far away from BS suffer from limited communication rates due to poor signal coverage and deteriorated channel quality. Besides, during large-scale emergencies, e.g. natural disasters, power outages and terrorist attacks, etc., BS becomes hard to access while the UE requests abruptly increase, making it difficult for the ground BS to accommodate all the communication and computing needs [5]. Given such limitations of MEC systems relying on terrestrial infrastructure, it is critical to create a platform capable of providing MEC service for UEs in remote areas or during emergencies [6].

As the cost of manufacturing UAVs continues to drop and the miniaturization of communication equipment continues to advance, it is feasible to install an MEC server on a UAV and establish wireless communication to perform computation-intensive tasks for ground-based UEs [7]. Unlike MEC systems relying on ground infrastructure, UAV-enabled MEC systems offer enhanced cost-effectiveness and enable quicker deployment, along with increased flexibility for reconfiguration, making them ideal for specific scenarios such as emergency response and rescue. The UAV can fly over the UEs, increasing the probability of establishing a line-of-sight (LoS) connection with the UEs. Moreover, the UAV is highly mobile but also controllable, which allows dynamic adjustment of its three-dimensional position and leads to significantly reduced transmission delay and lower energy consumption [8]. Besides serving as an airborne MEC server, UAV can also function as a relay to improve the communication capability of remote UEs and help them reach the ground MEC server [2], [9], [10]. Then, a more flexible architecture lets the UAV flexibly switch between the roles of an MEC server and a relay according to the dynamics of the environment to satisfy the QoE of the UEs. This air-ground coordinated architecture

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is beneficial to some practical applications. For example, the UEs for maritime operations are distant from the coast BS and the UAV can either provide task processing service or relay the task to the coast BS for processing [11]. Another example is the scenario of disaster relief, where pictures and videos captured by ground UEs need to be offloaded for disaster monitoring and assessment, while nearby BS may be destroyed or unavailable. In this situation, the UAV can either serve as the MEC server or as a relay to further offload the task to available BS to meet the service needs.

In fact, such an air-ground coordination architecture can also cope with the energy supply issue for the UAV. Recently, energy harvesting (EH) and wireless power transfer (WPT) technologies have been incorporated into UAV-enabled MEC systems to ensure sustainable energy supply [12], yet in practice, the efficacy of radio frequency (RF) energy harvesting may degrade due to path loss in less favorable geographical location and environmental conditions [9], [13]. A more power-sufficient laser-charged WPT technology has been incorporated into UAV-enabled architectures to improve the harvested energy of the UAV [14]. With a narrower energy laser beam, the laser power receiver can obtain hundreds of watts of power, and the field test has confirmed the feasibility of a laser-charged UAV [15]. Fortunately, in an air-ground coordination architecture, the laser emitter can co-locate with the ground MEC server to transfer wireless power to the UAV and provide sustained energy.

Although the UAV can harvest energy from the laser beam, the amount of harvested energy is uncertain. Thus, it becomes essential to utilize this energy efficiently to improve communication performance and extend the endurance of the UAV. [16]. Motivated by the aforementioned facts, in this paper, we consider an air-ground coordinated MEC system, in which the UAV is powered by the laser beam emitted from ground AP and it can simultaneously serve as a MEC server handling tasks from UEs and as a relay offloading tasks from UEs to the ground AP for processing. We aim to minimize the energy consumption of the UAV by jointly optimizing the UAV's trajectory, task and time allocation. The presented work has the following major contributions:

- Based on an air-ground coordinated MEC system, we formulate a problem to minimize the long-term average energy consumption of the UAV by jointly planning the UAV's trajectory, the task allocation for computation on UAV and computation on AP relayed via UAV, and UAV's time allocation for energy harvesting.
- Given the complexity and non-convex nature of the formulated problem due to coupled optimization variables, a two-step alternating optimization algorithm has been developed. This algorithm aims to decouple the optimization variables, enabling the problem to be solved iteratively by addressing two distinct subproblems in alternating steps, where subproblem 1 regarding computation task allocation and EH time allocation is a linear programming (LP) problem and is solved by standard tools such as PULP [17], while subproblem 2 regarding UAV's real-time trajectory scheduling is solved by a deep reinforcement learning (DRL) method, i.e., deep deterministic policy gradient (DDPG).
- Extensive simulations have been conducted to verify the effectiveness of our proposed two-step alternating algorithm LP-DDPG. Simulation results show that the convergence of LP-DDPG can be guaranteed and it exhibits the lowest UAV energy consumption compared with other benchmark algorithms under various scenarios. Features of the optimal policy regarding the UAV's trajectory and task allocation ratio are also demonstrated and discussed.

The rest of this paper is organized as follows. Section II states related works. Section III introduces system models and formulates the optimization problem. The proposed algorithm is detailed in Section IV. Simulation results are demonstrated in Section V, followed by conclusions in Section VI.

II. Related Works

In this section, we briefly review state-of-the-art works that are highly related to our work regarding UAV-enabled, air-ground coordinated and WPT-assisted MEC systems, respectively. We also review some widely utilized methods to solve related optimization problems and present the motivation of our work.

A. UAV-Enabled MEC

In recent years, the research community has dedicated extensive efforts to UAV-enabled MEC systems for different scenarios and optimization objectives, such as minimizing the system energy consumption [18], [19], minimizing the service latency [20], maximizing the service rate [21], maximizing the average security computation capacity [22], maximizing the UAV's efficiency [23], etc. The work in [18] minimized the total energy consumption for all UEs via jointly optimizing UAVs' movement, resource allocation and user association. In [19], a reconfigurable intelligent surface (RIS) was integrated into a non-orthogonal multiple access (NOMA)-based UAV-MEC system to further improve the network spectrum efficiency and the authors formulated a UAV energy consumption minimization problem by simultaneously optimizing both UAV's movement and the beamforming matrix at the RIS. The work in [20] aimed at minimizing the service latency of all tasks with constraints on learning accuracy, task processing latency and energy consumption, via jointly optimizing UAV's movement, computation and communication resource allocation, and the DNN model decision. In [21], Haber et al. aimed to maximize the rate of incoming offloading requests, while ensuring the UAVs' available energy levels, the required reliability and the latency of tasks. They achieved the goal by optimizing the allocated computational and radio resources, the UEs' associations, and the UAVs' movement. Mission redundancy and UAV failure rates were considered to avoid service outages. This work offered valuable insights into optimization strategies for maintaining ultra-reliable low-latency communication (URLLC) services. In [22], a NOMA-based UAV-MEC system with a flying eavesdropper was considered. The authors aimed to maximize the system's average secure computing capacity while ensuring a minimum secure computing requirement for each UE. They achieved the goal by jointly optimizing the movement of the UAV, local computation, transmit power, CPU computation frequency and the channel relationship coefficient allocation. In [23], Lin et al. tried to maximize the energy efficiency of the UAV while ensuring offloading fairness among UEs by jointly optimizing the flight time and movement of the UAV, binary offloading decisions and allocated time for each UE.

All the above-mentioned studies consider UAV as the sole MEC server for ground UEs. However, compared with BS and AP, UAV has limited service capability and may struggle to provide sufficient computing service.

B. Air-Ground Coordinated MEC

The air-ground coordinated MEC framework has been proven to effectively deal with the following two cases, i.e., the ground infrastructure cannot directly provide reliable connections for UEs and the UAV resource is limited when there are a large number of UEs accessing. Existing literature investigating air-ground coordinated MEC considered both single UAV scenarios [24], [25] and scenarios involving multiple UAVs [5], [26], [27]. In [24], Liu et al. investigated an MEC network with a multi-antenna UAV and aimed to minimize system energy consumption via jointly optimizing the beamforming vector, UAV's movement, the CPU frequency, and the transmit power and CPU frequency for the UEs. Their solution exhibited excellent performance especially when the computation tasks were intensive. In [25], Hu et al. aimed to minimize the combined energy consumption of both UAV and UEs, considering constraints of UEs' tasks, the information-causality, UAV's velocity and bandwidth allocation, by jointly optimizing UAV's movement, bandwidth allocation and computation resource scheduling. The work in [5] proposed a heterogeneous aerial-ground MEC framework with multiple UAVs. To improve the offloading efficiency, a partial offloading scheme and a time-division multiple access (TDMA) were adopted. A total computation bits maximization problem was formulated by jointly optimizing UAVs' movement, the transmit power and offloading time of both UEs and UAVs, the CPU frequencies of UEs and computing UAVs, as well as the UEs' association. In [26], Liu et al. investigated a MEC network assisted by multiple UAVs with air-ground cooperation, offering computing services for UEs with limited resources, where trajectory control, offloading proportion, the association and resource allocation on both computation and communication were jointly optimized to minimize the weighted energy consumption. A similar hybrid UAV-and-BS enabled MEC system was considered in [27]. To maximize the lifetime of UEs, Dai et al. proposed to minimize the energy consumption of all UEs by jointly optimizing the offloading decisions, the UAVs' movement and the allocation of computation and communication resources. Their solution greatly reduced the energy consumption of UEs and prolonged the lifespan of the system.

C. WPT-Assisted MEC

Recent advances in WPT technology have enabled WPT-assisted MEC for sustainable service provision. Through investigation, it is found that current work mostly focuses on WPT for ground UEs [28], [29], [30] with very few considerations of supplying wireless power for UAVs [9]. This is mainly because UAV requires greater power support than ground UEs for its operation. In [28], Du et al. considered UEs harvested energy from the UAV utilizing WPT technology and transmitted data to the UAV in a TDMA fashion. They aimed to minimize the UAV's energy consumption by jointly optimizing wireless powering duration, allocation of computing resources, the hovering time of the UAV, UEs association, the uplink transmission rate and the service sequence of the UEs, which resulted in great enhancement of energy efficiency compared with conventional UAV-enabled system. In [29], a UAV was dispatched to wirelessly charge multiple UEs using WPT technology and subsequently the UEs utilized the harvested energy to transmit the sensed information back to the UAV. The problem of minimizing the average age of information (AoI) of the UEs was investigated via jointly optimizing UAV's movement and the scheduling of information transmission and energy harvesting of UEs. In [30], a directional beamforming scheme was adopted for WPT. Considering the fairness issue, the authors proposed a problem with the goal of maximizing the minimum harvested energy across all UEs over a fixed time period by jointly optimizing the orientation of the directional antenna on the UAV and the trajectory planning of the UAV. The work in [9] was the first that assumed both UAV and UEs were charged by WPT technology, where the UAV was powered by laser beams emitted from AP and UEs were powered by UAV leveraging RF charging. The authors aimed to maximize the weighted sum of UEs' completed amount of tasks via jointly optimizing time and task allocation, UAV's movement and transmit power.

D. Related Optimization Technology

From the above-mentioned works, we have found that, once UAV is involved in the system, UAV positions often require dynamic adjustments over time, and corresponding parameters need to be adjusted accordingly. Thus, the optimization variables are time-related with high dimensionality. Furthermore, most of the formulated optimization problems are mix-integer, non-convex problems, which are intractable to solve. A natural way of solving such a complex problem is decomposing the problem into several sub-problems and solving them iteratively. In [5], Hu et al. decomposed the formulated problem into three distinct sub-problems: the UEs' offloading decisions, the resource allocation and the movement of the UAV. An efficient two-layered alternative optimization algorithm was introduced to approximate the optimal solution. Techniques such as Taylor expansion, convex optimization and successive convex approximation (SCA) were jointly utilized. In [24], the original problem was decomposed into three manageable sub-problems. To address these sub-problems, a three-stage alternating algorithm was introduced, involving the fixation of certain optimization variables while optimizing others. The solution process incorporated a combination of the SCA method, derived rank-one solution, sub-gradient method, and Lagrangian duality method. Similarly in [9], Hu et al. applied a three-step block coordinate descent

TABLE I
COMPARISON BETWEEN OUR WORK AND THE EXISTING WORKS

Ref.	Consider collaboration of UAV and AP	Consider WPT	Optimization objective	Decompose the problem	Non-learning method	DRL method
[18]	✗	✗	Min. UEs' energy consumption	✓	✓	✓
[19]	✗	✗	Min. UAV's energy consumption	✗	✗	✓
[20]	✗	✗	Min. the latency	✓	✓	✗
[21]	✗	✗	Max. the rate of the offloading requests	✓	✓	✗
[22]	✗	✗	Max. the average security computation capacity	✓	✓	✗
[23]	✗	✗	Max. UAV's energy efficiency	✓	✗	✓
[24]	✓	✗	Min. the weighted sum energy consumption of UEs and UAV	✓	✓	✗
[25]	✓	✗	Min. the weighted sum energy consumption of UEs and UAV	✓	✓	✗
[26]	✓	✗	Min. the weighted sum energy consumption of UEs and UAV	✗	✗	✓
[27]	✓	✗	Max. UEs' lifetime	✓	✗	✓
[28]	✗	✓	Min. UAV's energy consumption	✓	✓	✗
[29]	✗	✓	Min. the average AoI of UEs	✗	✗	✓
[30]	✗	✓	Max. the minimum harvested energy among UEs	✗	✓	✗
Our work	✓	✓	Min. UAV's energy consumption	✓	✗	✓

(BCD) algorithm to iteratively tackle three sub-problems and obtained a feasible solution. Aside from traditional optimization schemes, DRL-based algorithms have recently been used as a promising approach for effectively learning the optimal policy in dynamic environments. In [2], a cooperative multi-agent deep reinforcement learning (MADRL) framework was investigated to derive the policy including offloading decision, trajectory planning, and power control. The twin delayed deep deterministic policy gradient (TD3) algorithm was adopted to deal with the high-dimensional continuous action space, which showed its superiority over other optimization methods. In [26], a DRL approach utilizing multi-agent proximal policy optimization (MAPPO) was proposed to train both UEs and UAVs for cooperative decision-making in dynamic MEC networks. Attention mechanisms and beta distribution were introduced in critic and actor networks respectively to further enhance the training performance. Despite the good performance of some MADRL methods on the optimization problem with high-dimensional action space, for scenarios with a large number of UAVs or UEs, the state and action space may grow exponentially, resulting in lower convergence efficiency. In view of this, some work decomposed the formulated problem into sub-problems and solved one of the sub-problems via the DRL method [18], [31], [32], which led to easier convergence because of dimensionality reduction. In [18], the original problem was decomposed into two sub-problems, with one addressing the trajectory of the UAV, and the other handling resource allocation and user association. To facilitate real-time decision-making, a DRL-based trajectory planning algorithm was developed. Then, the authors introduced a low-complexity matching algorithm to optimize offloading decision and resource allocation. In [31], the multi-agent deep deterministic policy gradient (MADDPG) method was adopted to independently control the movement of each UAV. With the UAVs' trajectories determined, a low-complexity method was proposed to optimize UEs' offloading decisions. In [32], the problem of minimizing the UEs' AoI in reconfigurable intelligent surface (RIS)-assisted UAV system was decomposed into two sub-problems, where a soft actor-critic (SAC) algorithm was developed to learn policies of offloading decision and UAV's movement, and an alternating optimization algorithm was applied to address RIS phase shifts.

E. Motivation

Inspired by the above work, we also consider the novel air-ground coordinated system, where the UAV serves as both a mobile computing server for UEs and a relay for the ground AP. Such a system also facilitates the UAV to be wirelessly powered by the laser beam emitted from the ground AP. In such a system, due to the uncertainty of WPT, we aim to minimize the long-term average energy consumption of the UAV under the constraints of UEs' mission completion delay. To achieve this, UAV's trajectory, computation task allocation and EH time allocation need to be optimized.

Our work differs from previous literature in several aspects. We summarize the differences in Table I. Firstly, it involves collaborative task processing between UAV and AP, necessitating optimized task allocation. Secondly, the UAV harvests energy from the AP's laser beam, requiring optimization of the EH time and data transmission time allocation. Thirdly, the UAV requires real-time trajectory adjustments. All these decisions vary with external states such as channel conditions, task demands, and the UAV's previous position. On the one hand, traditional non-learning optimization methods are not suitable for real-time dynamic scenarios since they cannot capture the dynamic characteristics and time dependencies of the system. On the other hand, although our decision variables are continuous, the dimensionality of the variables becomes high when the number of UEs is large. Directly applying DRL or MADRL methods may struggle to converge. Previous literature has addressed this issue by employing problem decomposition, but different problems require different decomposition techniques. For our formulated problem, we have thoroughly analyzed its characteristics and found that if keeping the UAV's position fixed, the optimization problem concerning the ratios of the task allocation and time allocation is a strictly linear problem.

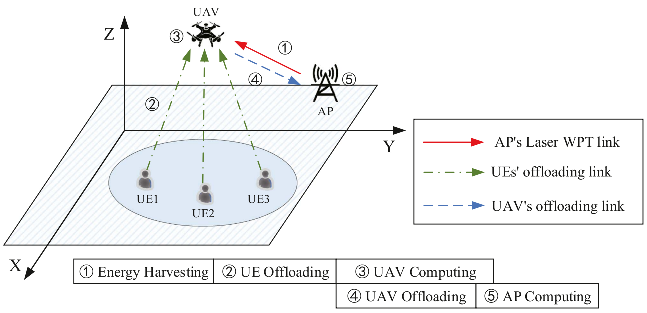


Fig. 1. UAV-assisted wireless powered MEC system architecture.

Thus we decompose the original problem into two subproblems and utilize a standard tool PULP to solve subproblem 1. For the other subproblem of optimizing UAV's trajectory, considering the dynamic environment and continuous action space, we adopted the DDPG algorithm to solve it. By iteratively solving two subproblems, we simplify the solution complexity, thereby enhancing the algorithm's performance.

III. System Model and Problem Formulation

Fig. 1 illustrates the focus of this paper: a laser-powered air-ground coordinated MEC system. It features one rotary-wing UAV equipped with an MEC server, a set of N different UEs denoted as $\mathcal{N} = \{1, 2,..., N\}$ and an AP. While all these nodes have a single antenna, the three types of nodes differ significantly in capacities. Specifically, the UEs cannot perform local computing, while the UAV is equipped with circuits for wireless EH and communication as well as the computing processors with limited computational capabilities. On the contrary, the grid-powered AP has a high-performance processing server with high-speed transmission and computation. Moreover, the AP is equipped with a laser transmitter that can generate high-power energy beams toward the UAV for energy provision. The whole system follows a harvest-then-offload workflow within each time slot, as illustrated by the numbered lines in Fig. 1: ① energy harvesting from AP to UAV, ② transmission from UEs to UAV, ③ computation at the UAV, ④ transmission from UAV to AP, ⑤ computation at the AP. Not only does the UAV accept UE offloads (step ②) and process tasks as an MEC server (step ③), it also serves as a relay, offloading tasks to the AP (step ④) for computation (step ⑤), and all these steps run in a time-share fashion. The system time is divided into T equal time slots with equal length τ, i.e. $t \in \mathcal{T}$, $\mathcal{T} = \{1, 2, \ldots, T\}$. Within a time slot, the UAV's position remains unchanged. This assumption is equivalent to discretizing the trajectory of the UAV based on the duration of time slots, which facilitates subsequent modeling and analysis, and is also widely adopted in the literature [25], [27] and [29]. Additionally, we assume a block-fading channel model, where the channel state remains constant within each block (time slot) and changes independently from one block to another. This model is commonly used in the analysis and simulation of wireless communication systems [22], [24], [26], [33], allowing for simplified modeling while still capturing the time-varying nature of the channel. In each time slot t, UE n will periodically generate a task $L_n(t) = \{D_n(t), C_n, T_{\max}\}$, where $D_n(t)$ is the size of task data, C_n is the number of cycles required to process unit data and $T_{\max}$ (i.e., time slot length τ) is the maximum tolerated delay.

A. UAV Movement

We consider a three-dimensional Cartesian coordinate system. Assume that the UAV's flying altitude Z remains constant, and the coordinates of the UAV are represented as $\omega(t) = [X(t), Y(t), Z]^T$, where $X(t), Y(t), Z$ are the spatial coordinates of UAV at time slot t, respectively. Also, assume that the angle of horizontal flight for the UAV in time slot t is

$$0 \leq \theta(t) \leq 2\pi, \tag{1}$$

and the distance as

$$0 \leq d(t) \leq d_{\max}, \tag{2}$$

where $d_{\max}$ is the UAV's maximal flying distance within a time slot. We assume that the UAV maintains a constant velocity $V(t)$ within time slot t, which reflects on the flying distance $d(t)$

$$V(t) = \frac{d(t)}{\tau}. \tag{3}$$

Additionally, the movements of the UAV are subject to the following constraints so that it remains within the rectangle-shaped service area,

$$0 \leq X(t) \leq X_{\max}, \tag{4}$$

$$0 \leq Y(t) \leq Y_{\max}, \tag{5}$$

where $X_{\max}$ and $Y_{\max}$ are the side lengths of the rectangle-shaped area, respectively. We also assume that UAV has a minimum elevation angle $\phi_{\min}$. Thus, the maximum coverage of the UAV $R_{\max}$ can be calculated as

$$R_{\max} = \frac{Z}{\tan(\phi_{\min})}. \tag{6}$$

When any UE is within this coverage limit $R_{\max}$, it can offload computational tasks to the UAV. We denote $\alpha_n(t) \in \{0, 1\}$ as a binary offloading decision variable of UE n, where $\alpha_n(t) = 1$ means that the task generated by UE n at time slot t is offloaded to the UAV for execution, and otherwise, $\alpha_n(t) = 0$. Furthermore, let $N(t)$ be the number of UEs served by UAV in time slot t.

B. UAV Propulsion Energy

The primary energy consumption during UAV flight is attributed to propulsion consumption, surpassing the energy consumed for communication and computing. Similar to [16], we utilize an existing analytical model for the propulsion power $P(V(t))$ of a rotary-wing UAV:

$$P(V(t)) = \underbrace{c_1\left(1 + \frac{3V(t)^2}{c_2^2}\right)}_{\text{blade profile}} + \underbrace{c_3\left(\sqrt{c_4 + \frac{V(t)^4}{4}} - \frac{V(t)^2}{2}\right)^{\frac{1}{2}}}_{\text{induced}} + \underbrace{c_5 V(t)^3}_{\text{parasite}}, \tag{7}$$

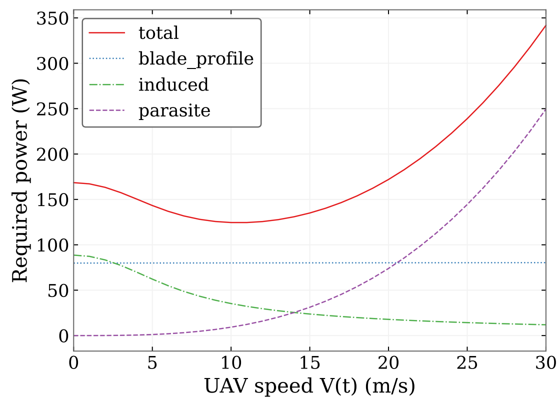


Fig. 2. The relationship between the propulsion power and UAV speed.

where c_1, c_2, c_3, c_4 and c_5 represent the constants associated with the hardware of the UAV and its aerodynamic parameters.

This analytical model covers both the hovering and flying modes of a UAV, and defines three components of $P(V(t))$, namely blade profile power, induced power, and parasite power, respectively. The blade profile power increases quadratically with $V(t)$, while the parasite power increases cubically with $V(t)$, and the induced power decreases with $V(t)$. Fig. 2 shows a typical diagram of how $P(V(t))$ and its three components change with UAV flight speed $V(t)$. With $V(t) = 0$, we get the power consumption of UAV in hover, Noticeably, as $V(t)$ increases, $P(V(t))$ decreases initially and then increases with $V(t)$, indicating that hovering does not represent the most power-conserving status overall.

Thus, the UAV's propulsion energy consumption at time slot t is

$$E_u^{prop}(t) = P(V(t))\tau. \tag{8}$$

C. Task Execution

Fig. 1 illustrates the process of task offloading for the entire system: 1) Task offloading from UEs to UAV; 2) Computation at the UAV; 3) Task offloading from UAV to AP; 4) Computation at the AP. Note that due to the small size of computation results, we do not consider the delay and energy consumption associated with sending the task results back to the UEs. Furthermore, to prevent interference among the UEs during the task offloading, we adopt the orthogonal frequency-division multiplexing (OFDM) protocol. Besides, we assume that the UAV can obtain the precise locations of all the UEs via regular handshaking attempts performed by the UEs towards the UAV.

1) Task Offloading From UEs to UAV: At each time slot, all the UEs within the coverage of the UAV will offload the task to it. We denote $\omega_n = [X_n, Y_n, 0]$ as the location of UE n in the 3D Cartesian coordinate system, and the Euclidean distance between UAV and UE n can be given by

$$d_n(t) = \|\omega(t) - \omega_n\|. \tag{9}$$

Let $h_n(t)$ denote the channel gain between UE n and the UAV. $h_n(t)$ involves both LoS and non-LoS (NLoS) links [34], and is given by:

$$h_n(t) = P_{n,Los}(t)\frac{g_0}{[d_n(t)]^2} + (1 - P_{n,Los}(t))\frac{\varepsilon g_0}{[d_n(t)]^2}, \tag{10}$$

where g_0 represents the reference channel gain at the distance of 1 m, and $\varepsilon < 1$ is the NLoS additional attenuation factor. $P_{n,Los}(t)$ denotes LoS probability between the UAV and UE n at time slot t, and we assume that $P_{n,Los}(t)$ remains unchanged within each time slot. In general, the probability of LoS propagation depends on the surrounding environment and the elevation angle, which can be approximated using a modified sigmoid function:

$$P_{n,Los}(t) = \frac{1}{1 + xe^{-y(\phi_n(t) - x)}}, \tag{11}$$

where x and y are parameters depending on the surrounding environment (e.g., rural, suburban and urban), and $\phi_n(t)$ is the elevation angle between the UAV and the UE n, which is

$$\phi_n(t) = \frac{180}{\pi}\arctan\left(\frac{Z}{\sqrt{(X(t) - X_n)^2 + (Y(t) - Y_n)^2}}\right). \tag{12}$$

Assume that uplink bandwidth B is evenly allocated to each UE during task offloading. Thus, the transmission rate between UE n and UAV is given by

$$R_n(t) = \frac{B}{N(t)}\log\left(1 + \frac{h_n(t)P_n}{\sigma_u^2}\right), \tag{13}$$

where P_n is the transmit power of UE n, σ_u^2 is the noise power consumption by the UAV.

Thus, the transmission delay between UE n and UAV as well as the UAV's energy consumption for receiving tasks are given by

$$T_{n,u}^O(t) = \frac{D_n(t)}{R_n(t)}, \tag{14}$$

and

$$E_{n,u}^O(t) = P_u^r T_{n,u}^O(t) = \frac{P_u^r D_n(t)}{R_n(t)}, \tag{15}$$

respectively, where P_u^r is the receiving power of UAV.

2) Computation at the UAV: After the UEs offload the task to the UAV, the UAV will decide the proportion of tasks to be computed locally and further offload the remaining tasks to the AP through the UAV relay. Therefore, the computation delay and energy consumption for local execution at UAV are given by

$$T_u^C(t) = \frac{\beta_n^u(t)D_n(t)C_n}{f_n^u(t)}, \tag{16}$$

and

$$E_u^C(t) = \kappa(f_n^u(t))^3 T_n^C(t) = \kappa(f_n^u(t))^2\beta_n^u(t)D_n(t)C_n, \tag{17}$$

respectively, where $\beta_n^u(t) \in [0, 1]$ denotes the fraction of tasks of UE n executed at UAV. $\beta_n^u(t) = 1$ and $\beta_n^u(t) = 0$ means all tasks of UE n are computed at UAV and AP, respectively. $f_n^u(t)$ is the computation resource allocated by UAV for handling tasks from UE n. Assuming that the total computation resource F_u

of the UAV is equally allocated to each served UE, we have $f_n^u(t) = \frac{F_u}{N(t)}$. Finally, $\kappa > 0$ indicates the effective switched capacitance.

3) Task Offloading From UAV to AP: We denote $\omega_{AP} = [X_{AP}, Y_{AP}, 0]$ as the fixed location of AP in the 3D Cartesian coordinate system, and the distance between UAV and AP in time slot t is given by

$$d_u(t) = \|\omega(t) - \omega_{AP}\|. \tag{18}$$

Considering the scenario that the UAV relays some UEs' tasks to the AP, the channel gain between the UAV and the AP is given by

$$h_u(t) = \frac{g_0}{[d_u(t)]^2}, \tag{19}$$

and the transmission rate between UAV and AP is given by

$$R_u(t) = B_u \log\left(1 + \frac{h_u(t)P_u}{\sigma_{AP}^2}\right), \tag{20}$$

where B_u is the bandwidth, P_u denotes the transmit power of UAV and σ_{AP}^2 is the noise power at AP. Given $R_u(t)$, the transmission delay between UAV and AP, and the UAVs' energy consumption for offloading tasks are

$$T_{u,AP}^O(t) = \frac{\beta_n^{AP}(t)D_n(t)}{R_u(t)}, \tag{21}$$

and

$$E_{u,AP}^O(t) = P_u T_{u,AP}^O(t) = \frac{P_u \beta_n^{AP}(t)D_n(t)}{R_u(t)}, \tag{22}$$

respectively, where $\beta_n^{AP}(t) \in [0,1]$ denotes the proportion of tasks from UE n to be executed at AP and satisfies $\beta_n^u(t) + \beta_n^{AP}(t) = 1$.

4) Computation At the AP: After receiving the task from the UAV, the AP will handle the computation task, and the computation delay at the AP is given by

$$T_{AP}^C(t) = \frac{\beta_n^{AP}(t)D_n(t)C_n}{f_{AP}^u(t)}, \tag{23}$$

where $f_{AP}^u(t)$ is the computation resource allocated by AP for tasks from UE n. Similarly, the total computation resource F_{AP} of the AP is equally allocated to each served UE, thus we have $f_{AP}^u(t) = \frac{F_{AP}}{N_{AP}(t)}$, with $N_{AP}(t)$ being the number of UEs served by the AP in time slot t.

Note that the assumption of equal resource allocation among UEs served by the UAV and AP simplifies the model by providing a straightforward approach to resource management. However, in more dynamic scenarios, this simple assumption may not always yield optimal outcomes. Adaptive resource allocation strategies can be employed to achieve improved performance and resource utilization. These strategies involve dynamic allocation of resources based on channel conditions and UE priorities. In fact, there are numerous variables that can be adjusted to optimize system performance, but we cannot cover all aspects comprehensively. We can only strike a balance between performance and complexity. Considering the above reasons and relevant literature [2], we assume in our research that the bandwidth and computing resources are evenly allocated.

D. Energy Harvesting

The energy of the UAV is harvested from the AP via laser charging. At the beginning of each time slot, the AP charges the UAV for $q(t)\tau$ time, where $q(t) \in [0,1]$. Similar to [9], we use the following linear EH model to quantify the harvested energy in time slot t,

$$E_u^{EH}(t) = \eta_u P_{AP} g_{AP}(t) q(t)\tau, \tag{24}$$

where $\eta_u \in [0,1]$ is the energy conversion efficiency, P_{AP} is the transmit power of AP, and $g_{AP}(t) = \frac{\Lambda \vartheta e^{-\alpha d_u(t)}}{(D+\zeta d_u(t))^2}$ is the equivalent channel power gain for laser charing from AP to UAV, where Λ, ϑ, D and ζ are all constants depending on the laser transmitter and receiver hardware, and α is the attenuation coefficient. Furthermore, We assume that the radio is on during the EH phase, and thus the energy consumption of UAV during the EH phase is given by

$$E_u^{con}(t) = P_u^r q(t)\tau. \tag{25}$$

E. Problem Formulation

In slot t, once all tasks of all UEs are completed, the overall energy consumption of UAV is given by

$$\begin{aligned} E_u(t) = &\sum_{n=1}^{N} \alpha_n(t)[E_{n,u}^O(t) + E_u^C(t) + E_{u,AP}^O(t)] \\ &+ E_u^{con}(t) + E_u^{prop}(t). \end{aligned} \tag{26}$$

The constraint in (27) ensures that the UAV always harvests enough energy, i.e., within each time slot the harvested energy is no less than the sum of energy consumed for all purposes:

$$E_u(t) \leq E_u^{EH}(t). \tag{27}$$

In addition, we assume that the UAV has communication and computation modules running in parallel. Specifically, the UAV can simultaneously carry out local task computing and task transmission to AP. Therefore, the total completion time for UAV to process UE n's task is given by

$$T_n(t) = q(t)\tau + T_{n,u}^O(t) + \max\{T_u^C(t), T_{u,AP}^O(t) + T_{AP}^C(t)\}. \tag{28}$$

Moreover, it takes time for the UAV to harvest energy via WPT, and faster energy harvesting creates a better chance for the system to meet the constraints on task completion delay.

Summarizing the models above for the air-ground coordinated MEC system, the problem of minimizing the long-term average energy consumption of the UAV can be formulated by jointly optimizing the UAV's position $\omega(t)$, computation task allocation ($\beta_n^u(t)$ and $\beta_n^{AP}(t)$), and EH time ratio $q(t)$. Specifically, let $\boldsymbol{U} = \{\omega(t), t \in \mathcal{T}\}$, $\boldsymbol{\beta} = \{\beta_n^u(t), \beta_n^{AP}(t), \forall n \in \mathcal{N}, t \in \mathcal{T}\}$, $\boldsymbol{Q} = \{q(t), t \in \mathcal{T}\}$. The formulated problem is described as:

$$\mathcal{P}1: \min_{\boldsymbol{U},\boldsymbol{\beta},\boldsymbol{Q}} \lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} E_u(t) \tag{29a}$$

$$\text{s.t. } 0 \leq \theta(t) \leq 2\pi, t \in \mathcal{T} \tag{29b}$$

$$0 \leq d(t) \leq d_{\max}, t \in \mathcal{T} \tag{29c}$$

$$0 \le X(t) \le X_{\max}, 0 \le Y(t) \le Y_{\max}, t \in \mathcal{T} \tag{29d}$$

$$0 \le \beta_n^u(t) \le 1, \forall n \in \mathcal{N}, t \in \mathcal{T} \tag{29e}$$

$$0 \le \beta_n^{AP}(t) \le 1, \forall n \in \mathcal{N}, t \in \mathcal{T} \tag{29f}$$

$$\beta_n^u(t) + \beta_n^{AP}(t) = 1, \forall n \in \mathcal{N}, t \in \mathcal{T} \tag{29g}$$

$$0 \le q(t) \le 1, t \in \mathcal{T} \tag{29h}$$

$$T_n(t) \le T_{\max}, \forall n \in \mathcal{N}, t \in \mathcal{T} \tag{29i}$$

$$E_u(t) \le E_u^{EH}(t), t \in \mathcal{T} \tag{29j}$$

where (29b)–(29d) define the constraints on the UAV's movement, (29e)–(29g) are constraints for two types of task offloading, (29h) is the constraint about the EH time of the UAV, (29i) ensures that the UAV always completes the computation of any UE task within the maximum tolerated delay, (29j) guarantees that the UAV never consumes more energy than captured.

Problem $\mathcal{P}1$ is a non-convex problem. Moreover, computation task and EH time allocations are strongly coupled with the UAV's trajectory, and the UAVs' movement may influence both task execution delay and energy consumption, especially when the network environment is unstable. Thus, it is infeasible to obtain the optimal solution via traditional optimization schemes. In the next section, we propose a two-step alternating optimization framework to address the challenges.

IV. Proposed Solution LP-DDPG

Our framework addresses the joint optimization problem in two steps. In step 1, we optimize allocations of computation tasks and EH time, i.e., the variables $\{\boldsymbol{\beta}, \boldsymbol{Q}\}$ with fixed UAV's trajectory $\{\boldsymbol{U}\}$. In Step 2, we optimize the UAV's trajectory $\{\boldsymbol{U}\}$ with fixed $\{\boldsymbol{\beta}, \boldsymbol{Q}\}$.

A. Step 1: Optimizing $\{\boldsymbol{\beta}, \boldsymbol{Q}\}$ With Fixed $\boldsymbol{U}$

Once UAV's trajectory is determined, the original problem $\mathcal{P}1$ can be simplified as a static problem within the current time slot:

$$\mathcal{P}2 : \min_{\boldsymbol{\beta}, \boldsymbol{Q}} E_u^{con}(t) + \sum_{n=1}^{N} \alpha_n(t)[E_u^C(t) + E_{u,AP}^O(t)]$$
$$\text{s.t. (29e)} - \text{(29j)} \tag{30}$$

Problem $\mathcal{P}2$ is a typical linear programming problem, as all variables in $\boldsymbol{\beta}$ and $\boldsymbol{Q}$ are continuous and all of its constraints are linear. Thus, we use the standard Python package PULP to solve $\mathcal{P}2$.

B. Step 2: Optimizing $\boldsymbol{U}$ With Fixed $\{\boldsymbol{\beta}, \boldsymbol{Q}\}$

Given fixed computation task and EH time allocations, the problem of UAV's trajectory optimization is formulated as

$$\mathcal{P}3 : \min_{\boldsymbol{U}} \lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \{E_u^{prop}(t) + \sum_{n=1}^{N} \alpha_n(t)[E_{n,u}^O(t) + E_{u,AP}^O(t)]\}$$
$$\text{s.t. (29b)} - \text{(29d)}, \text{(29i)}, \text{(29j)} \tag{31}$$

Problem $\mathcal{P}3$ is still non-convex, yet the dynamic real-time environment, especially those related to emergencies, creates high demands on fast decision-making regarding the changing of UAV trajectory. Conventional numerical optimization methods are unsuitable as they typically demand a substantial number of iterations to achieve satisfactory results, which leads to excessive computational complexity. Therefore, we propose a DRL-based approach to solve the real-time scheduling of UAV trajectory defined in $\mathcal{P}3$. Specifically, the DDPG algorithm [35] is suitable for problems with continuous variables like the UAV's movement. When performing actions, DDPG adheres to the principles of the DPG theorem [36]. Unlike traditional stochastic policy-gradient algorithms producing a list of actions with generated probabilities, DDPG deterministically takes actions according to a parameterized policy, thus DDPG does not require integration of the whole action space. Combining neural networks and actor-critic architecture, DDPG demonstrates improved efficiency in addressing problems with high-dimensional action spaces.

1) MDP Formulation: Here, we re-model problem $\mathcal{P}3$ as a Markov Decision Process (MDP). In our system, the UAV's energy consumption is determined by the actions of UAV and the current state of the system environment. Meanwhile, the action of UAV will impact the current system environment and trigger a new random state. Thus, problem $\mathcal{P}3$ can be formulated as an MDP $(\mathcal{S}, \mathcal{A}, \mathcal{R})$, where $\mathcal{S}$ represents the state space, $\mathcal{A}$ represents the action space, and $\mathcal{R}$ is the reward function.

State space $\mathcal{S}$: According to the UAV trajectory optimization problem, the state $s(t)$ consists of the UAV's 3D coordinate position, each UE's task size and the horizontal distance $d_n^H(t)$ between each UE n and the UAV, that is,

$$s(t) = \{\omega(t), D_n(t), d_n^H(t), \forall n \in \mathcal{N}\}. \tag{32}$$

Action space $\mathcal{A}$: the space for the UAV's action is composed of the horizontal direction angle $\theta(t) \in [0, 2\pi]$ and the horizontal flying distance $d(t) \in [0, d_{\max}]$, i.e.,

$$a(t) = \{\theta(t), d(t)\}. \tag{33}$$

Reward function $\mathcal{R}$: Technically, the reward function should conform with the objective of the optimization problem. As the objective of problem $\mathcal{P}3$ is to minimize the UAV's energy consumption, the reward function should be set as the negative of energy consumption. Nevertheless, a successful solution to problem $\mathcal{P}3$ is also required to satisfy certain constraints. Therefore, we incorporate constraints into the reward function and define different reward functions for different scenarios.

$$\mathcal{R}(t) = \begin{cases} R_1(t), & \text{if N(t) = 0} \\ R_2(t), & \text{otherwise,} \end{cases} \tag{34}$$

$$R_1(t) = -\omega_1 \sum_{n=1}^{N} d_n^H(t) - \omega_2 E_u(t) + p\Gamma(29d), \tag{35}$$

$$R_2(t) = \omega_3 \sum_{n=1}^{N} \alpha_n(t) D_n(t) - \omega_4 E_u(t) + p\Gamma(29d), \tag{36}$$

Specifically, as the UAV needs to cover as many UEs as possible, we incorporate the negative sum of the horizontal distances

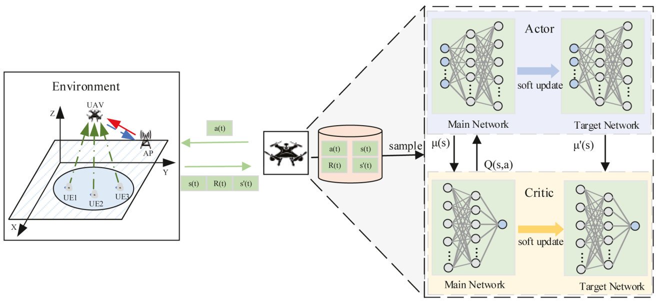


Fig. 3. The structure of DDPG algorithm.

between UEs and the UAV into the reward function, i.e., the first term in (35), under the condition that none of the UEs is covered by the UAV ($N(t) = 0$). Otherwise, when the UAV covers at least one UE, we add the task volume of the UAV as an additional reward, see the first term in (36). Furthermore, for violation condition (29d), we add a penalty term Γ(29d), where $\Gamma(\cdot)$ is an indicator function, whose value is 0 if condition (29d) is met, otherwise, it is -1. Coefficients $\omega 1, \omega 2, \omega 3, \omega 4$ and p are positive constants balancing the reward and the penalty terms.

Fig. 3 illustrates the framework of the DDPG approach to solve the above MDP for the UAV trajectory in the context of the high-dimensional continuous action space. In the DDPG algorithm, an action $a(t)$ are generated through the actor network and a critic network is employed to generate the Q-value evaluating the action produced by the actor network. In addition, the DDPG adopts a target actor network with weights $\theta^{\pi'}$ and a target critic network with weights $\theta^{Q'}$ to improve the learning stability.

C. Summary of Proposed Solution LP-DDPG

Algorithm 1 summarizes our proposed solution LP-DDPG for computation offloading and UAV trajectory control. Firstly, we initialize the parameters of the four neural networks and the replay buffer $\mathcal{B}$, corresponding to line 1–3. At the beginning of each episode, the UAV obtains the state from the environment. Then, it selects an action based on the actor network $\pi(s(t)|\theta^{\pi})$. In addition, a random noise ξ is added to the action to enhance exploration, where ξ follows a normal distribution, and we normalize the processed result to the range of [0,1]. In line 8, given the UAV's current position, we use PULP to obtain the optimal computation task and EH time allocations. After that, the UAV executes the movements. Then, after the action is taken, the UAV receives reward $R(t)$ according to (34) and observes the updated system state $s(t+1)$. To enhance sample efficiency and stabilize the training process, the UAV stores the transition $(s(t), a(t), R(t), s(t+1))$ in the replay buffer $\mathcal{B}$. When the buffer is full, it samples a random mini-batch of transitions with size $\mathcal{M}$ from $\mathcal{B}$. Then, UAV updates the critic network by using the loss function, $L(\theta^Q)$, that is,

$$L(\theta^Q) = \frac{1}{\mathcal{M}} \sum_{m=1}^{\mathcal{M}} [y_m - Q(s(m), a(m)|\theta^Q)]^2, \tag{37}$$

where the target values y_m can be obtained by

$$y_m = r(m) + \gamma Q(s(m+1), a(m+1)|\theta^Q). \tag{38}$$

Algorithm 1: DDPG-based Approach for Computation Offloading and Trajectory Control (LP-DDPG).

1: Initialize actor network $\pi(s(t)|\theta^{\pi})$ with weights θ^{π} and critic network $Q(s(t), a(t)|\theta^Q)$ with weights θ^Q;
2: Initialize target networks $\pi'(\cdot)$ with weights $\theta^{\pi'} = \theta^{\pi}$ and $Q'(\cdot)$ with weights $\theta^{Q'} = \theta^Q$;
3: Initialize experience replay buffer $\mathcal{B}$
4: **for** episode = 1,2,...,$e^{\max}$ **do**
5: Initialize the state $s(t)$
6: **for** time slot t = 1,2,...,T **do**
7: UAV selects action $a(t) = \pi(s(t)|\theta^{\pi}) + \xi$;
8: Solve problem ($\mathcal{P}2$) using PULP to obtain the optimal computation task allocation $(\beta_n^u(t), \beta_n^{AP}(t))$ and EH time ratio $q(t)$, given UAV's current position $\omega(t)$;
9: UAV executes action $a(t)$ with the given $\beta_n^u(t)$, $\beta_n^{AP}(t)$ and $q(t)$;
10: According to (34), UAV obtains the immediate reward $R(t)$ and then enters the next state $s(t+1)$;
11: Store transition $(s(t), a(t), R(t), s(t+1))$ into $\mathcal{B}$
12: $s(t) \leftarrow s(t+1)$;
13: **if** the replay buffer is full **then**
14: Sample a random mini-batch of $\mathcal{M}$ transitions $(s(t), a(t), R(t), s(t+1))$ from $\mathcal{B}$;
15: According to (37), update weights of the critic network θ^Q;
16: According to (39), update weights of the actor network θ^{π};
17: According to (40) and (41), Soft update the two target networks.
18: **end if**
19: **end for**
20: **end for**

Additionally, the actor network can be trained by the policy gradient defined as

$$\nabla_{\theta^{\pi}} J \approx \frac{1}{\mathcal{M}} \sum_{m=1}^{\mathcal{M}} \nabla_a Q(s, a|\theta^Q)|_{s=s(m), a=\pi(s(m)|\theta^{\pi})}$$
$$= \frac{1}{\mathcal{M}} \sum_{m=1}^{\mathcal{M}} [\nabla_a Q(s, a|\theta^Q)|_{s=s(m), a=\pi(s(m))} \cdot \nabla_{\theta^{\pi}} \pi(s|\theta^{\pi})|_{s=s(m)}]. \tag{39}$$

Finally, according to (37) and (39), the target networks can be trained by

$$\theta^{Q'} \leftarrow \lambda \theta^Q + (1-\lambda)\theta^{Q'}, \tag{40}$$
$$\theta^{\pi'} \leftarrow \lambda \theta^{\pi} + (1-\lambda)\theta^{\pi'}, \tag{41}$$

where λ is the updating rate.

D. Computation Complexity

The complexity of Algorithm 1 depends on the strategies for solving the two subproblems. In step 1, The complexity of

TABLE II
SIMULATION PARAMETERS

Parameters	Value	Parameters	Value
Flight altitude of the UAV Z [2]	100 m	Transmit power of UE P_n [2]	0.1 W
Maximum flight distance $d_{\max}$ [6]	10 m	Computation resource F_u [26]	50 GHz
Size of input data D_n [2]	[5,8] Mbits	Computation resource F_{AP}	80 GHz
Number of CPU cycles C_n [2]	500 cycles/bit	Effective switched capacitance κ [2]	10^{-28}
Maximum tolerated delay $T_{\max}$	0.5 s	Noise power δ_u^2, δ_{AP}^2 [2]	-100 dBm
Minimum elevation angle $\phi_{\min}$ [2]	$\frac{\pi}{4}$	Energy conversion efficiency η_u [9]	0.5
Reference channel gain g_0 [2]	-50 dB	Attenuation coefficient α [9]	10^{-6} m^{-1}
Environment parameters x and y [6]	10, 0.6	Weight coefficient ω_1	2×10^{-5}
NLoS attenuation ε [6]	0.2	Weight coefficient ω_2	0.0001
Uplink channel bandwidth B [9]	300 MHz	Weight coefficient ω_3	0.01
Bandwidth preassigned to AP B_u [26]	10 MHz	Weight coefficient ω_4	0.001
Transmit power of UAV P_u [26]	0.2 W	Penalty coefficient p	0.1
Receiving power of UAV P_u^r [2]	0.2 W		

$\mathcal{P}2$ is primarily dictated by the overall number of optimization variables. Specifically, for a system with N UEs and running for T time slots, the overall number of variables to optimize is $(N+1)T$. As we choose the solver based on the interior-point method (IPM) for the optimization of the computation task allocation and EH time ratio, the complexity of solving $\mathcal{P}2$ is $\mathcal{O}((NT+T)^{3.5}\log(1/\epsilon))$, where $\epsilon > 0$ indicates the solution accuracy.

Step 2 focuses on using the DDPG algorithm to solve the UAV trajectory optimization problem $\mathcal{P}3$, which is a non-convex problem in nature. The computational complexity of a neural network is commonly quantified by the number of operations involved in the network model, which is determined by factors such as the dimensions of input states and actions, the number of neurons in each layer, and the overall number of neural network layers. For our DDPG framework shown in Fig. 3, the computational complexity must take into account both the actor and critic networks. The layer counts in the actor and critic neural networks are denoted as L_a and L_c, respectively, and for the l-th layer, n_l^a and n_l^c signify the quantities of neurons in the actor and critic networks, respectively. Thus, the complexity of the DDPG algorithm is $\mathcal{O}(e^{\max}T(\sum_{l=1}^{L_a} n_{l-1}^a n_l^a + \sum_{l=1}^{L_c} n_{l-1}^c n_l^c))$.

Therefore, the overall complexity of the DDPG-TP algorithm is $\mathcal{O}(e^{\max}((NT+T)^{3.5}\log(1/\epsilon) + T(\sum_{l=1}^{L_a} n_{l-1}^a n_l^a + \sum_{l=1}^{L_c} n_{l-1}^c n_l^c)))$. However, it is worth mentioning that the training process does not need to occur on the UAV. Instead, it can be implemented on the cloud server with greater computational capacity and no energy restrictions. Once the algorithm converges, the trained model can be deployed on the UAV. The UAV will then gather state information, input them into the model for inference, and obtain the corresponding actions. The computational complexity and energy consumption of this inference process are both manageable.

V. SIMULATION RESULTS

In this section, we report the performance of our proposed LP-DDPG algorithm in a series of simulations. As illustrated in Fig. 4, we consider an air-ground coordinated MEC system in a rectangle-shaped area of 400×400 m^2, and it features 10 UEs distributed randomly within a hotspot area with a radius of 75 m. Unless otherwise stated, Table II summarizes the settings for the main simulation parameters. The simulation platform features Intel(R) Core(TM) i7-10750H CPU, NVIDIA GeForce GTX 1650Ti GPU as well as software developed in Python 3.6, PULP 2.6.0, and Tensorflow 2.6.2. The actor and critic networks in the LP-DDPG algorithm share the same network structure and training process, i.e., each neural network contains two hidden layers with 128 neurons in total, and is trained by using AdamOptimzer with a learning rate of 0.001. Moreover, we set the discount factor $\gamma = 0.9$, and the batch size $\mathcal{M} = 100$. Each simulation training epoch contains 100 time slots with a length τ of 0.5 s for each slot.

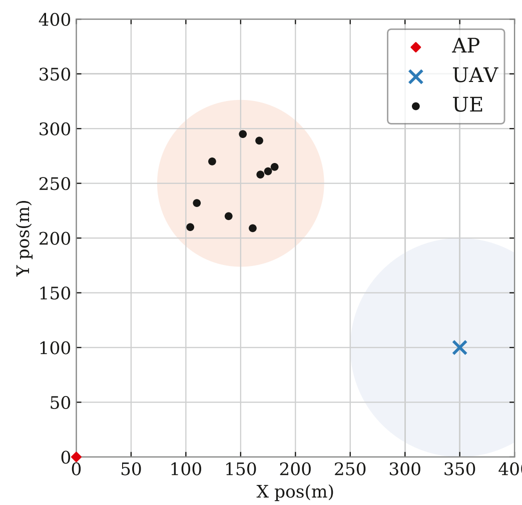


Fig. 4. Locations of 10 UEs, an AP and a UAV in the UAV-enabled wireless powered MEC system for simulations.

The objective of our work is to minimize the long-term average energy consumption of UAV by jointly optimizing the allocation ratio of computing tasks, the energy harvest time ratio, and the UAV trajectory. Thus, in developing benchmark testing algorithms, we assess the influence of different strategies on UAV energy consumption and task completion rate. First, we explore different task offloading strategies, such as complete offloading [2], average offloading and random offloading [27]. Then, we consider a different flight mode, i.e., fixed trajectory of the UAV [5], [10]. Additionally, since we decompose the

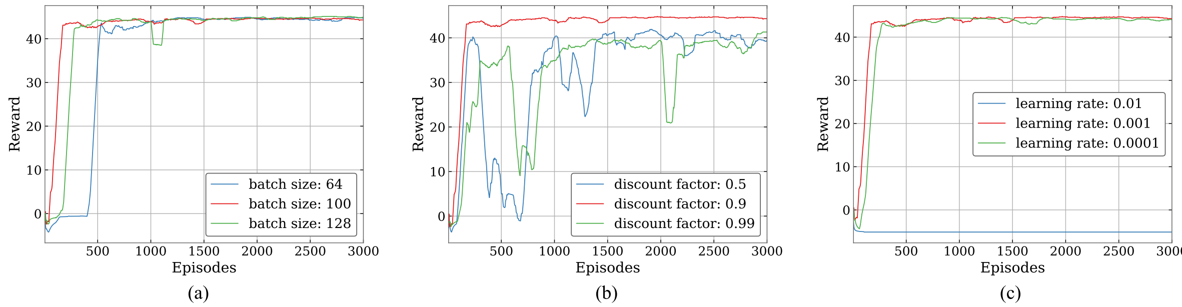


Fig. 5. Convergence performance of LP-DDPG algorithm under different parameter settings. (a) Batch size. (b) Discount factor. (c) Learning rate.

original problem into two subproblems and solve them using LP and DDPG algorithms, respectively, to evaluate the efficiency of the DDPG algorithm, we also test another DRL algorithm PPO, which has been effectively applied in the literature [26], as a comparative approach for addressing the second subproblem. The results are generated based on some system models we introduce in Section III. For example, the total energy consumption of the UAV in each time slot is computed based on (26), and the task completion rate is measured by dividing the number of tasks successfully computed within the maximum tolerable delay by the total number of tasks.

In summary, we compare our proposed LP-DDPG algorithm with the following five benchmark schemes:

- *Complete Offloading (CO) scheme:* The UAV serves as a mobile relay only, i.e., the UAV only transmits all UEs' offloading tasks to the AP for processing.
- *Average Offloading (AO) scheme:* Given the distribution range of the task sizes, the UAV manages computing tasks that fall below the average, while the AP tackles those that exceed the average.
- *Random Offloading (RO) scheme:* The proportions of tasks offloaded by the UEs to the UAV and the AP are randomly assigned.
- *Fixed Trajectory (FT) scheme:* The UAV remains stationary at the center of all UEs throughout each time slot, providing them with computing and offloading services.
- $\mathbf{PPO-basedScheme(LP-PPO):}$ Instead of DDPG, we use another DRL method PPO to solve the UAV trajectory optimization problem.

A. Hyperparameter Comparison

In this subsection, we compare the convergence performance of the LP-DDPG algorithm under different hyperparameter settings during the training process. Specifically, Fig. 5(a) shows that with different values of batch size, LP-DDPG always converges and tends to stabilize as the learning proceeds with more episodes. However, the fastest convergence speed is achieved with a batch size of 100. In Fig. 5(b), it can be seen that with the discount factors of 0.5 and 0.99, there are significant fluctuations in the learning process, leading to a slow convergence rate. When the discount factor is 0.9, convergence occurs relatively quickly. As shown in Fig. 5(c), when the learning rates of both the actor and critic networks are set to 0.01, the algorithm fails to converge. However, convergence is achieved within 300 episodes with rates of 0.001 and 0.0001, with the convergence rate slightly faster when set to 0.001. Therefore, in the following simulation, we set the default values of the batch size, the discount factor and the learning rate to 100, 0.9 and 0.001, respectively.

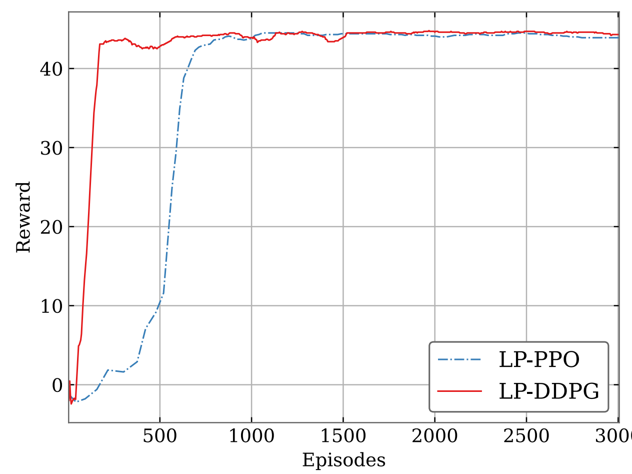
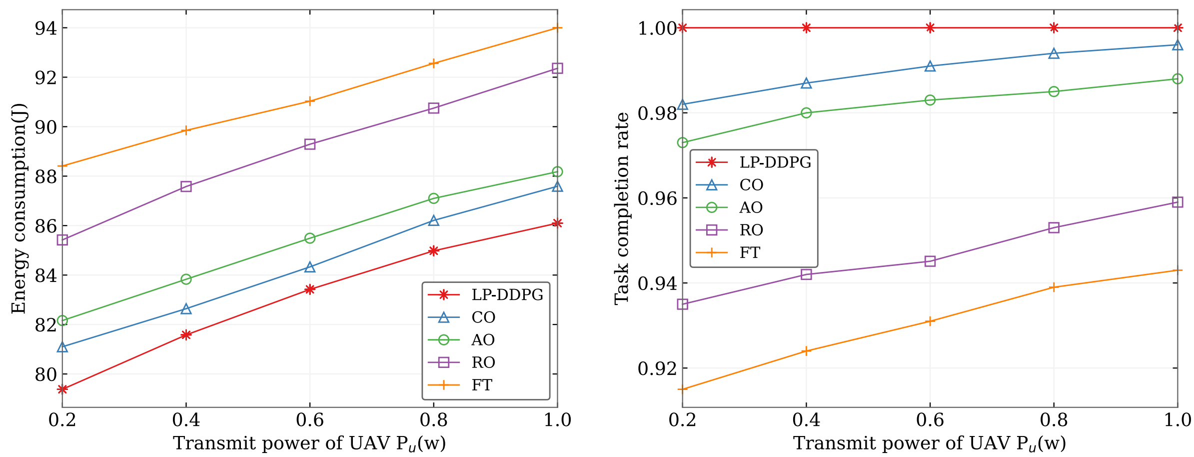


Fig. 6. The convergence performance of LP-DDPG and LP-PPO.

B. Performance Comparison of Different Algorithms

In this subsection, we compare our proposed LP-DDPG algorithm with the aforementioned benchmark algorithms under different parameter settings.

Firstly, we analyze the convergence performance of LP-DDPG and LP-PPO algorithms. As shown in Fig. 6, the final convergence values of LP-DDPG algorithm and LP-PPO algorithm are almost identical. However, the LP-DDPG algorithm converges faster, reaching convergence in 170 episodes, whereas LP-PPO algorithm takes 800 episodes to converge. The difference between LP-DDPG and LP-PPO algorithms lies in the DRL algorithms adopted. Both DDPG and PPO are DRL algorithms suited for continuous action spaces. However, they differ in algorithm type and policy update methods. DDPG belongs to the class of actor-critic methods and uses a deterministic policy, where the actor approximates the optimal policy directly by outputting the best-believed action for each state, while PPO is an on-policy algorithm that employs stochastic policies and utilizing a clipped surrogate objective to update policies, ensuring that the policy updates do not deviate too far from the previous policy. Due to the characteristics of PPO,



(a) (b)

Fig. 7. Performance comparison under different power levels. (a) Energy consumption. (b) Task completion rate.

it exhibits less fluctuation during the convergence process, as shown in Fig. 6, but its convergence speed is slower than DDPG. In subsequent comparative experiments, we find that the performances achieved by LP-DDPG and LP-PPO algorithms are almost identical. Therefore, the performances of LP-PPO algorithm are not shown in other experiments.

Secondly, Fig. 7 illustrates the impact of UAV's transmit power on the UAV energy consumption and task completion rate derived by different algorithms. In Fig. 7(a), as P_u rises from 0.2W to 1.0W, the UAV energy consumption of all the five algorithms almost exhibits linear growth as expected. Due to its fixed trajectory design, the FT algorithm suffers the highest energy consumption. This highlights the importance of trajectory design in enhancing UAV's energy efficiency. The RO scheme has much higher energy consumption than CO scheme. This implies that an unreasonable task allocation is even worse than having the UAV not serve as an MEC server. Due to the fact that the AO algorithm requires offloading all tasks below the average value to the UAV for computation, its computational energy consumption exceeds that of offloading energy consumption. Therefore, the energy consumption generated by this algorithm is greater than that of the CO algorithm. The proposed LP-DDPG algorithm achieves the lowest energy consumption in all situations due to its joint optimization of trajectory, time and task allocation. Its energy savings over CO, AO, RO and FT schemes are 1.52%, 2.63%, 6.72% and 8.86% on average, and 2.21%, 3.38%, 7.07% and 10.21% at most, respectively. In Fig. 7(b), the task completion rate of our proposed algorithm remains at 100% under different transmission power levels, while the task completion rates of the other four comparison schemes gradually increase with the increase in transmission power. FT scheme achieves the lowest task completion rate. This is because the task completion delay depends on the delay of energy harvesting and task computation offloading. Although this scheme optimizes the task allocation ratio, the energy consumption generated when the UAV hovers is greater than that during its flight, requiring more time for energy harvesting to support task computation, thereby increasing the task completion delay. The task completion rate of the RO scheme is also relatively low because unreasonable task allocation increases the computation and offloading delays of the UAV, making it unable to complete all tasks within the maximum tolerated delay. Compared with the above two schemes, CO and AO schemes show higher task completion rates, both reaching more than 97%, but still lower than LP-DDPG algorithm.

Thirdly, the impact of the maximum tolerated delay $T_{\max}$ on the UAV energy consumption and task completion rate is shown in Fig. 8. In Fig. 8(a), it can be observed from all the curves that, with the increase in $T_{\max}$, the UAV energy consumption decreases. This is reasonable because when the delay constraint is tight, the UAV requires greater energy consumption to meet the requirement. Our proposed LP-DDPG algorithm still achieves the lowest energy consumption under all the $T_{\max}$ settings, with energy savings of 2.75%, 4.34%, 8.29% and 13.77% on average, and 4.14%, 6.20%, 10.68% and 20.74% at most over CO, AO, RO and FT schemes, respectively. It is worth noting that when $T_{\max} = 0.45$, the energy consumption of LP-DDPG is very close to that of the CO scheme, which implies that when the delay constraint is tight, optimal policy tends towards the UAV acting as a relay, offloading the UEs' tasks to the AP for computation. In Fig. 8(b), our proposed algorithm maintains a task completion rate of 100% under different maximum tolerable delays, while the task completion rates of the other four comparative schemes gradually increase with the increase in maximum tolerable delay, with the rate of increase gradually slowing down and tending toward 100%. This is because the relaxation of the maximum tolerable delay constraint allows UAV and AP more ample time to compute tasks for UEs, while the optimization space also becomes increasingly limited.

Fourthly, Fig. 9 shows the impact of the number of UEs N on the UAV energy consumption and task completion rate. In Fig. 9(a), with an increasing number of UEs accessing the system, the workload of UAV also rises, which leads to higher energy consumption in communication, computing, and propulsion. Therefore, the energy consumption derived by all four algorithms increases. Again, the proposed LP-DDPG algorithm outperforms all the other benchmark schemes. Compared with CO, AO, RO and FT schemes, it achieves energy savings of 2.07%, 3.61%, 8.04% and 11.87% on average, and 2.70%, 4.15%, 8.98% and 13.44% at most, respectively. In Fig. 9(b), under different number of UEs, the task completion rate of the

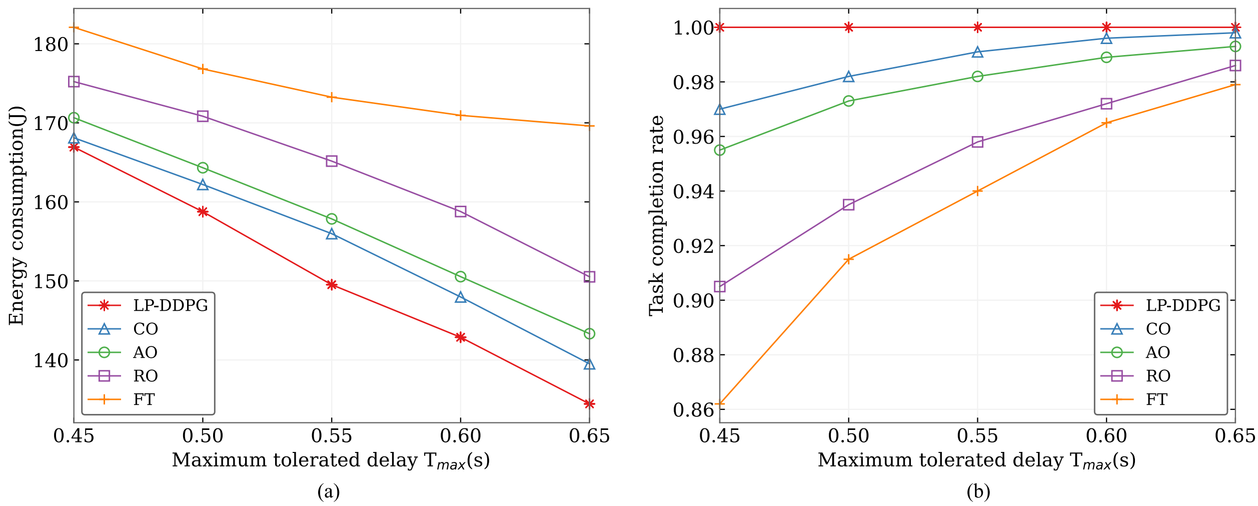


Fig. 8. Performance comparison under different maximum tolerated delays. (a) Energy consumption. (b) Task completion rate.

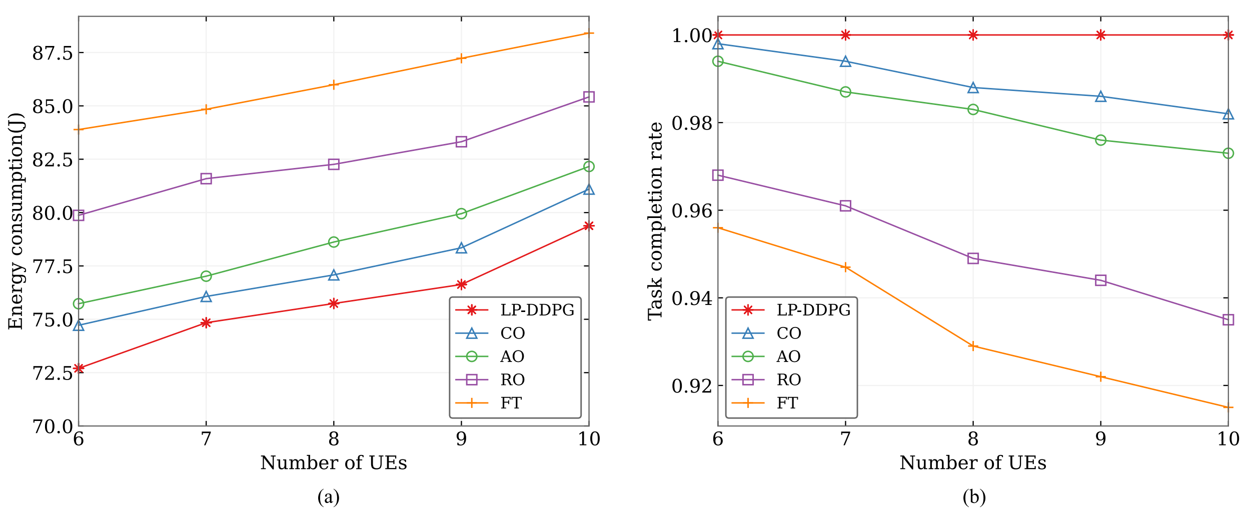


Fig. 9. Performance comparison under different numbers of UEs. (a) Energy consumption. (b) Task completion rate.

proposed algorithm remains at 100%, while the task completion rate of the other four comparison schemes decreases with the increase of the number of UEs. This is because with fewer UEs, there is a reduced workload. Consequently, the required latency for UAV and AP to handle tasks from UEs also decreases, which can better meet the constraint of the maximum tolerated delay. Similarly, the FT scheme performs the worst. The completion rates of tasks for the CO and AO schemes are close to 100% when the number of UEs is small. As the number of UEs increases, the completion rates gradually decrease but still outperform that achieved by the RO scheme.

C. Trajectory Evaluation and Task Allocation

To better understand the behavior pattern of the UAV trajectory and the task allocation strategy learned by our proposed LP-DDPG algorithm, Figs. 10 and 11 show the flight trajectory of the UAV and probability distribution of the proportion of tasks offloaded to UAV, respectively, under default

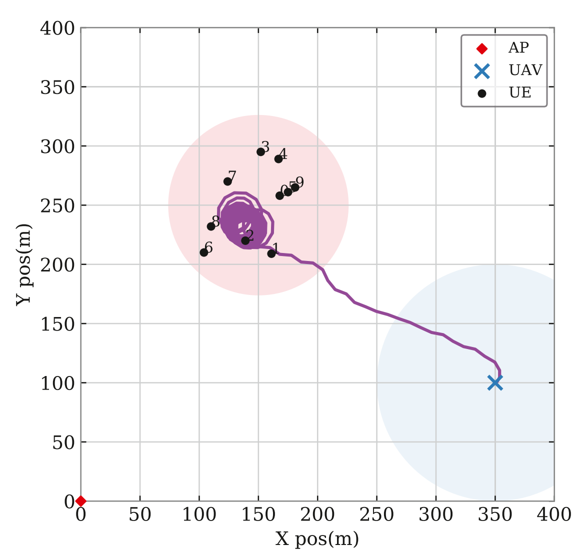


Fig. 10. The flight trajectory of UAV.

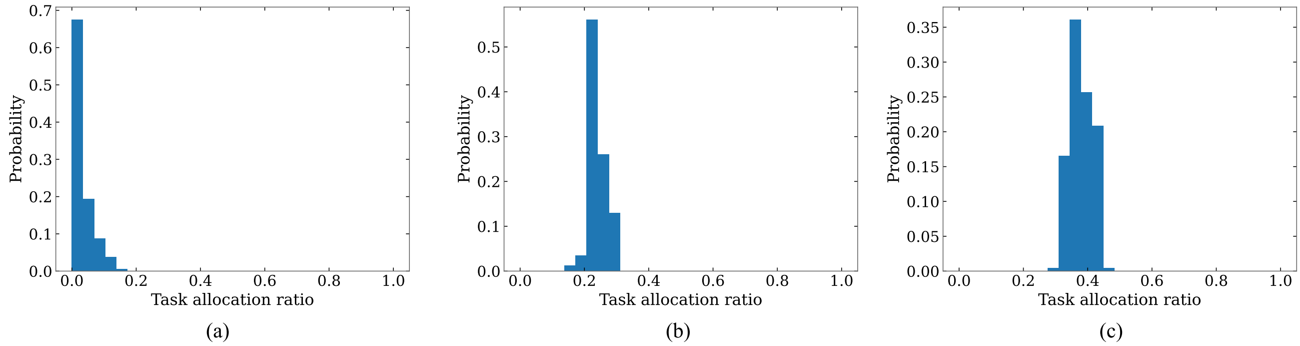


Fig. 11. Probability distributions of task allocation ratios under different task sizes, (a) task size: 5-6 Mbits, (b) task size: 6-7 Mbits, (c) task size: 7-8 Mbits.

settings. Both figures show the results obtained after running 600 time slots. In Fig. 10, the red diamond at the origin represents the AP, the blue cross represents the UAV and the black dots are the UEs. Additionally, the red shaded circle illustrates the hotspot area of UE deployment and the blue shaded one illustrates the coverage range of the UAV. It can be observed from Fig. 10 that the UAV first flies towards the hotspot area and then hovers over the UEs to provide MEC or relay service. It is worth noting that the UAV only hovers around a very small subarea within the hotspot area and this subarea is relatively closer to the AP. Possible reasons may include the following points: first, according to the coverage range of the UAV, hovering around the small subarea can effectively cover all the UEs with the least amount of movement distance, and thus UAV's energy can be saved; second, the small hovering area closer to the AP allows the UAV to capture more energy, enhance the UAV-to-AP transmission rate, thereby making it easier to meet the delay constraint.

As the size of each task is a random value in the range of [5, 8] Mbits according to the simulation setting, we divide the range of the task size into three intervals, i.e., [5, 6), [6, 7) and [7, 8]. Fig. 11 shows the probability distribution of the proportion of tasks offloaded to UAV for MEC under the three task size intervals. It can be seen from Fig. 11(a), 11(b), and 11(c) that the peak mode region is gradually shifting towards the right. That is, as the task size increases, the amount of tasks offloaded for computation on the UAV gradually increases. This is because when the task size is small, both the delay and energy consumption incurred by transmission to the AP are small. However, as the task size increases, transmission to the AP leads to longer delay and energy consumption and thus it is more efficient to have more tasks computed on the UAV.

Noting that DDPG itself exhibits some degree of generalization capability, due to the incorporation of batch normalization mechanism. However, the validation of our algorithm is still confined to simulation environments. Some models, such as the link model and energy model, may not accurately mirror real-world conditions. A more pragmatic approach is to gather state information in real-world scenarios for model training. Furthermore, deploying pre-trained models and refining them during deployment is another viable method.

VI. Conclusion

This paper investigates an air-ground coordinated MEC system with a UAV and an AP collaboratively relaying and processing offloaded computation tasks from UEs. An optimization problem is formulated to achieve minimum long-term average energy consumption of the UAV by jointly designing the UAV's trajectory, computation task and EH time allocation. A two-step alternating algorithm LP-DDPG is proposed to tackle this non-convex problem, by deposing the original problem into two sub-problems. LP method is used to solve subproblem 1 regarding computation task and EH time allocation, while the DDPG-based method is proposed to solve subproblem 2 regarding UAV's real-time trajectory design. Extensive simulation results show that the proposed LP-DDPG algorithm significantly reduces the average UAV energy consumption compared with other benchmark schemes. Future research will further consider incorporating the impact of adverse weather on UAV flight status and energy consumption in the system model, as well as incorporating a battery model to consider a hybrid energy supply for the UAV, enhancing the energy supply reliability.

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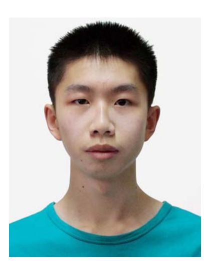

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